

- Q.6. (a) What is the "inert-pair effect"? Discuss the origins of the inert-pair effect. [1,3]
 (b) How are ionization energy and electron affinity related to 'absolute electronegativity' and 'chemical hardness'? [2]

Q.7. (a) Predict what type (if any) of molecular orbital would result from the combination of each pair of orbitals shown. The combining orbitals are all similar in energy. Assume the z-axis is the bonding axis in all cases. [4]

- i. $s + p_z$
- ii. $p_x + p_z$
- iii. $d_{xz} + d_{xz}$
- iv. $d_{xy} + d_{xy}$

} bonding orbitals

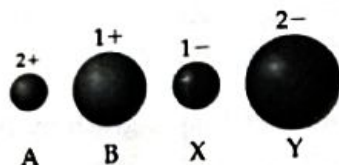
(b) Mention the three conditions must be met for choosing atomic orbitals (AOs) for linear combination and form MOs. [2]

Q.8. What is Born-Oppenheimer (BO) Approximation? Apply it in the construction of the Hamiltonian for the single electron in H_2^+ . Define or schematically show the parameters involved in the Hamiltonian. [3,3]

Q.9. In the following, four ions — A, B, X, and Y— are shown with their relative ionic radii. The ions shown in red carry positive charges: a 2+ charge for A and a 1+ charge for B. Ions shown in blue carry negative charges: a 1- charge for X and a 2- charge for Y.

(a) Which combinations of these ions produce ionic compounds where there is a 1:1 ratio of cations and anions? Why? [2]

(b) Among the combinations in part (a), which leads to the ionic compound having the largest lattice energy? Why? [4]



Q.10. (a) Name three factors on which the magnitude of the lattice energy of an ionic solid depends. Briefly explain how influence the lattice energy. [3]

(b) What is the main thermodynamic driving force for the thermite reaction and why does it arise? Explain. [3]

Given:

$$E_{p,min} = -A \frac{N_A |z_A z_B| e^2}{4\pi\epsilon_0 d} \left(1 - \frac{d^*}{d}\right)$$

General and Structural Chemistry
Mid-Semester Examination: Spring 2025
IIIT-Hyderabad

Time: 1:30 Hours

Full Marks: 60

Q.1. (a) Is each set of quantum numbers allowed? Explain your answers. [3]

- (i) $n = 2; l = 1; m_l = 2; m_s = +1/2$
- (ii) $n = 3, l = 0; m_l = -1; m_s = -1/2$
- (iii) $n = 2; l = 2; m_l = 1; m_s = +1/2$

(b) In a multi-electron atom, how do the energies of s, p, d, and f orbitals of a given shell compare? Justify your answer. [3]

Q.2. (a) The periodic table is divided into blocks. Identify each block and explain the principle behind the divisions. Which quantum number distinguishes the horizontal rows? [2,1]

(b) Write down the ground state electron configuration of chromium (${}_{24}\text{Cr}$). Explain the deviation from the Aufbau principle. However, there is no case where an atom such as carbon is found with the ground-state configuration of $1s^2 2s^1 2p^3$. Why? [1.5,1.5]

Q.3. (a) Show a schematic plot of the variation of the effective potential energy of an electron in the hydrogen atom vs. radius for the $l = 0$ and $l \neq 0$ wavefunctions. Explain the nature of the curves. [4]

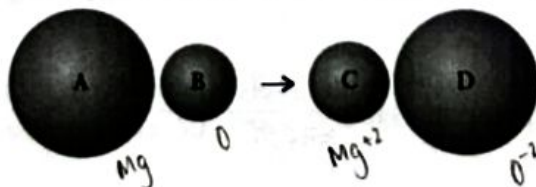
(b) The wavefunction of one of the d orbitals is proportional to $\cos \theta \sin \theta \cos \phi$. Identify the d-orbital. [2]

Q.4. (a) Define the radial distribution function. [1]

(b) For a 1s orbital, the radial distribution function is given by $P(r) = \frac{4Z^3}{a_0^3} r^2 e^{-2Zr/a_0}$. Find out the most probable location of an electron in an H atom in its ground state. [2]

(c) Calculate the most probable radius, r_{mp} , at which an electron will be found when it occupies a 1s orbital of He^+ . Compare the most probable radii of H and He^+ and comment on their values. (Given $a_0 = 52.9 \text{ pm}$). [3]

Q.5. (a) Below is pictured the reaction between an atom of oxygen and an atom of magnesium. Identify each element and the ions formed and explain your reasoning. [2]



(b) Write a balanced chemical equation for the reaction of solid selenium dioxide, $\text{SeO}_2(\text{s})$, with water. [2]

(c) Name two atoms whose size is smaller than fluorine. [2]